

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (CL/ME/EE/CE/IT/EC)

December 2012

CL 102.01 Environmental Sciences

Date: 13.12.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Define the following terms. [03]
a) Biotic b) Abiotic c) Biosphere.
- (b) Explain the objectives and guiding principles of Environmental Studies. [04]
- Q - 2 (a) Explain effect of air pollution on human beings. [04]
- (b) Write short note on Eutrophication. [05]
- (c) Explain the impact of technology and development on environment. [05]

OR

- Q - 2 (a) Explain "Acid Rain" as a major Environmental problem. [04]
- (b) List out all possible types of Environmental pollution and Explain the term environmental degradation and discuss the role of advanced technology in the degradation of environment. [05]
- (c) Explain with neat sketch temperature profile of atmosphere in detail. [05]
- Q - 3 (a) Explain general classification of air pollutants and describe classification according to sources. [04]
- (b) Define noise. Briefly describe the sources and effects of noise pollution. [05]
- (c) Enlist all current global environmental problems and discuss green house effect in details. [05]

OR

- Q - 3 (a) Enlist different air pollutant with their sources and effects. [04]
- (b) Enlist and explain sources of water pollution in detail. [05]
- (c) How depletion of Ozone layer does takes place in the atmosphere? Explain the ill effect of Ozone depletion. [05]

SECTION - II

- Q - 4 (a) Define 'Ecosystem'. Write objectives of ecological studies. [03]

- (b) Explain the functional and structural components of Ecosystem. [04]
- Q - 5 (a) Explain concept of food chain with example. [04]
- (b) Define "Bio- Geo- Chemical cycling". Explain "Sulphur cycle" with a neat sketch. [05]
- (c) What do you mean by Ecological Pyramids? Explain in detail "Pyramid of Number". [05]

OR

- Q - 5 (a) Discuss how Energy flow differs from Nutrient flow. [04]
- (b) Explain the process of Nitrogen fixation and Denitrification with neat sketch. [05]
- (c) Enlist Major Ecosystems. Explain in detail "Forest Ecosystem". [05]
- Q - 6 (a) Write a note on problems due to overexploitation of water resources. [04]
- (b) Differentiate between Conservation and Destruction of Natural Resources. [05]
- (c)
1. Define "Point Source Pollutant". [05]
 2. Define "Environmental Lapse Rate".
 3. Define "Chemical Oxygen Demand".
 4. Ozone is formed by _____ reaction.
 5. Total available air mass is _____ in nature.

OR

- Q - 6 (a) Write a note on "Geothermal Energy". [04]
- (b) Give advantages and disadvantages of Conventional energy resources. [05]
- (c)
1. Define "Qualitative Pollutants". [05]
 2. Define "Biodegradable Pollutants".
 3. As per EPA Environment is defined as _____.
 4. Zone consisting air, water and soil is known as _____.
 5. Inner core of lithosphere contains _____.
